Exploratory Data Analytics
(E1CSA231)

Micro Project Report

On

Exploratory Data Analysis of Weather History Dataset

Submitted by

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Table of Contents

Abstract

1. Introduction

    1.1 Project Overview

    1.2 Objectives of the Project

    1.3 Problem Statement

    1.4 Scope of the Project

2. Dataset Description

    2.1 Source of Data

    2.2 Dataset Overview

    2.3 Features

3. Data Preprocessing

4. Exploratory Data Analysis

5. Results and Insights

6. Conclusion

7. Future Scope

References

Appendix

Abstract

This project performs Exploratory Data Analysis (EDA) on the Weather History dataset to understand patterns in weather conditions. The dataset contains historical weather data including temperature, humidity, wind speed, pressure, and visibility collected over several years. The analysis was performed using Python, Pandas, Matplotlib, and Seaborn. Various statistical summaries and visualizations were used to identify trends and relationships between weather variables. The analysis shows patterns between temperature, humidity, and pressure, and highlights how environmental factors influence weather conditions.

1. Introduction

1.1 Project Overview

This project focuses on analysing a historical weather dataset using Exploratory Data Analysis techniques. The dataset contains different weather attributes such as temperature, humidity, wind speed, pressure, and visibility. By using statistical analysis and visualization techniques, patterns and relationships between these variables can be identified.

The project uses Python-based data analysis libraries to explore the dataset and generate insights from the data.

1.2 Objectives of the Project

- To analyze weather patterns using historical data
- To understand relationships between different weather variables
- To visualize distributions of temperature, humidity, and pressure
- Identifying correlations between environmental factors
- Generating meaningful insights from weather data

1.3 Problem Statement

Weather conditions are influenced by multiple environmental variables such as humidity, temperature, wind speed, and atmospheric pressure. Understanding the relationships between these factors is important for analyzing climate patterns.

This project aims to analyze historical weather data and identify patterns and relationships between different weather parameters.

1.4 Scope of the Project

The scope of this project includes:

- Data cleaning and preprocessing
- Statistical analysis of weather variables
- Visualization of weather patterns

This project focuses only on **data analysis and visualization**, not on building predictive models.

2. Dataset Description

2.1 Source of Data

The dataset used in this project is the **Weather History Dataset**, obtained from **Kaggle**.

2.2 Dataset Overview

- Total Rows: **96,453**
- Total Columns: **12**
- Dataset Type:
  - Numerical features (temperature, humidity, pressure, wind speed)
  - Categorical features (weather summary, precipitation type)

The dataset contains hourly weather information.

2.3 Features

Column Name	Description
Formatted Date	Date and time of weather record
Summary	General description of weather
Precip Type	Type of precipitation (rain/snow)
Temperature (C)	Temperature in Celsius
Apparent Temperature (C)	Feels-like temperature
Humidity	Relative humidity
Wind Speed (km/h)	Wind speed in kilometres per hour
Wind Bearing (degrees)	Wind direction in degrees
Visibility (km)	Visibility distance in kilometres
Cloud Cover	Cloud coverage in the sky
Pressure (millibars)	Atmospheric pressure
Daily Summary	Daily weather description

3. Data Preprocessing

3.1 Data Cleaning

The dataset was checked for duplicate records and inconsistencies. Any unnecessary or redundant data was removed to ensure data quality.

3.2 Handling Missing Values

The dataset was inspected for missing values. Any missing values were handled using appropriate techniques such as filling with mean values or removing incomplete rows where necessary.

3.3 Data Transformation

Several transformations were applied:

- Converting date column into datetime format
- Encoding categorical variables
- Standardizing column names for easier analysis

3.4 Outlier Detection

Outlier detection is an important step in data preprocessing because extreme values can significantly affect statistical analysis and visualization. In this project, outliers were identified primarily using **boxplots and basic statistical techniques**.

A **boxplot visualization** was used to detect unusual values in numerical variables such as **temperature, humidity, wind speed, pressure, and visibility**. In a boxplot, outliers appear as points that lie far outside the whiskers of the plot. These points represent values that differ significantly from most of the data.

In addition to visual inspection, **statistical methods such as the Interquartile Range (IQR)** were considered. The IQR measures the spread of the middle 50% of the data and helps determine whether a value is unusually high or low compared to the rest of the dataset. Values that fall below $Q1 - 1.5\ X\ IQR$ or above $Q3 + 1.5\ X\ IQR$are typically considered potential outliers.

During the analysis, some extreme values were observed in variables such as **wind speed and temperature**. These values were carefully reviewed to determine whether they represented errors or legitimate observations. Since the dataset represents **real-world weather measurements**, extreme values may correspond to actual weather events such as storms, strong winds, or unusual temperature conditions.

For this reason, the detected outliers were **not removed from the dataset**. Removing them could lead to loss of meaningful information about rare but important weather phenomena. Instead, the outliers were retained so that the dataset continues to reflect real environmental variations.

Keeping these values ensures that the analysis remains realistic and that the results accurately represent the full range of weather conditions present in the dataset.

4. Exploratory Data Analysis

4.1 Univariate Analysis

Single variable analysis was performed to understand distributions of weather parameters.
Examples:

- Temperature distribution using histograms
- Humidity distribution
- Wind speed distribution

These visualizations helped identify average conditions and extreme values.

4.2 Bivariate Analysis

Relationships between two variables were analyzed using scatter plots and bar charts.
Examples:

- Temperature vs Humidity
- Temperature vs Pressure
- Wind Speed vs Visibility

These comparisons help understand how different environmental factors influence each other**.**

4.3 Multivariate Analysis

Multivariate analysis involves studying **more than two variables at the same time** to understand how different factors interact with each other. In weather data, variables such as **temperature, humidity, wind speed, pressure, and visibility** are often interconnected, and analyzing them together helps reveal deeper patterns.

In this project, multivariate analysis was performed using **correlation heatmaps and pair plots**. A **correlation heatmap** was used to measure the strength of relationships between numerical variables. Correlation values range from **-1 to +1**, where +1 indicates a strong positive relationship, -1 indicates a strong negative relationship, and values close to 0 indicate little or no relationship.

The heatmap helped identify relationships among variables such as temperature, humidity, wind speed, pressure, and visibility. For example, temperature tends to have a strong relationship with apparent temperature, while humidity can influence visibility.

4.4 Data Visualization

Data visualization plays an important role in Exploratory Data Analysis because it helps present complex data in a clear and understandable form. Graphical representations make it easier to identify patterns, trends, and relationships within the dataset.

In this project, several types of visualizations were used to analyze the weather dataset. **Histograms** were used to study the distribution of numerical variables such as temperature, humidity, wind

speed, and pressure. These graphs help show how frequently different values occur and provide insight into the overall spread of the data.

Boxplots were used to detect outliers and understand the variation within the data. They show the median, quartiles, and extreme values, making it easier to identify unusual observations that may represent rare weather events.

Scatter plots were used to analyze relationships between two numerical variables. For example, plots such as temperature versus humidity and wind speed versus temperature help reveal whether any correlation exists between these variables.

In addition, **bar charts and count plots** were used for categorical variables such as precipitation type and weather summary. These charts display the frequency of each category and help understand which weather conditions occur most frequently..

4.5 Correlation Analysis

Correlation analysis is used to measure the **strength and direction of the relationship between numerical variables** in a dataset. In this project, a **correlation matrix** was created to analyze how different weather parameters are related to each other. The correlation values range from **-1 to +1**, where values closer to +1 indicate a strong positive relationship, values closer to -1 indicate a strong negative relationship, and values near 0 indicate little or no relationship.

To better understand these relationships, a **heatmap visualization** was used. The heatmap provides a color-coded representation of the correlation matrix, making it easier to identify strong or weak relationships between variables.

The analysis revealed several important relationships in the dataset. **Temperature and apparent temperature show a strong positive correlation**, which is expected because apparent temperature represents how hot or cold the temperature feels to humans. As the actual temperature increases, the apparent temperature generally increases as well.

Humidity shows a moderate relationship with both temperature and visibility. Higher humidity levels can often lead to lower visibility due to fog, mist, or moisture in the air. This indicates that humidity plays an important role in determining weather conditions.

On the other hand, **pressure shows a relatively weak correlation with most other variables** in the dataset. This suggests that atmospheric pressure may vary independently and may not have a direct strong relationship with variables such as temperature or wind speed.

Overall, correlation analysis helps identify which variables influence each other and provides valuable insights into how different weather factors interact within the dataset.

5. Results and Insights

This section summarizes the **most important findings obtained from the exploratory data analysis** of the weather dataset. By analyzing the visualizations and statistical relationships between variables, several meaningful patterns and trends were identified.

One of the most noticeable findings is that **temperature and apparent temperature show a strong positive correlation**. Apparent temperature represents how the temperature feels to humans after considering factors such as humidity and wind. Because of this, when the actual temperature increases, the apparent temperature also tends to increase. This strong relationship confirms the consistency and reliability of the dataset.

Another important observation is that **higher humidity levels are often associated with lower visibility**. When humidity increases, more moisture is present in the air, which can lead to conditions such as fog, mist, or haze. These conditions reduce how far objects can be clearly seen, which explains the negative relationship between humidity and visibility.

The analysis also showed that **wind speed varies significantly across different weather conditions**. While most wind speeds fall within a moderate range, there are occasional extreme values that indicate strong winds or storm-related events. These variations demonstrate how dynamic and unpredictable wind behavior can be in real-world weather systems.

From the temperature distribution analysis, it was observed that **most temperature values fall between 5°C and 20°C**. This suggests that the dataset primarily represents weather conditions within a moderate climate range rather than extreme hot or cold environments. The histogram visualization confirmed that temperature values are concentrated within this range.

In addition, **atmospheric pressure values remain relatively stable throughout the dataset**. Unlike temperature or humidity, pressure tends to fluctuate within a narrow range. This stability indicates that pressure changes are usually gradual and are often associated with larger atmospheric systems such as high-pressure or low-pressure regions.

The visualizations also revealed that **precipitation type influences temperature patterns**. For example, snow generally occurs at lower temperatures, while rain occurs at relatively higher temperatures. This observation aligns with natural weather behavior and validates the relationships present in the dataset.

Overall, the analysis highlights how **multiple environmental factors interact to influence weather conditions**. Variables such as temperature, humidity, wind speed, and pressure work together to shape daily weather patterns. These insights help improve our understanding of climate behavior and demonstrate the usefulness of exploratory data analysis in interpreting real-world datasets.

6. Conclusion

This project focused on performing **Exploratory Data Analysis (EDA)** on the Weather History dataset to better understand weather patterns and the relationships between different atmospheric variables. The dataset contained several important weather parameters such as temperature, humidity, wind speed, pressure, visibility, and precipitation type. By applying various EDA techniques, the dataset was explored in detail to identify patterns, trends, and relationships among these variables.

During the analysis, several preprocessing steps were performed such as **converting date formats, extracting useful time-based features like year and month, and examining the dataset for missing values and outliers**. These steps helped ensure that the dataset was clean and suitable for further analysis
.
Different types of visualizations such as **histograms, scatter plots, line plots, boxplots, and heatmaps** were used to better understand the data. These visual tools helped in identifying the distribution of weather variables, detecting unusual observations, and understanding how variables influence each other.

The results of the analysis revealed meaningful insights. For example, **temperature and apparent temperature showed a strong positive relationship**, while **humidity was found to affect visibility levels**. Wind speed showed variation across different weather conditions, and atmospheric pressure remained relatively stable across the dataset
.
Overall, the project demonstrates that **Exploratory Data Analysis is a powerful technique for understanding large and complex datasets**. By using statistical methods and visualization tools, it becomes easier to discover hidden patterns and relationships in the data.

The final outcome of this project is a **better understanding of environmental and weather-related patterns** present in the dataset. These insights can be useful for further studies such as climate analysis, weather prediction models, or advanced machine learning applications in meteorology.

7. Future Scope

Future improvements could include:

- Applying machine learning models for weather prediction
- Using larger or real-time weather datasets
- Performing time-series analysis on weather trends
- Developing dashboards for weather monitoring

References

1. Kaggle. Weather History Dataset. https://www.kaggle.com
2. McKinney, W. (2018). Python for Data Analysis. O'Reilly Media.
3. Pandas Documentation. https://pandas.pydata.org
4. Seaborn Visualization Library. https://seaborn.pydata.org

Appendix

PYTHON CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load dataset
df = pd.read_csv("weatherHistory (1).csv")
# Convert date column
df['Formatted Date'] = pd.to_datetime(df['Formatted Date'], utc=True)

# Extract useful features
df['Year'] = df['Formatted Date'].dt.year
df['Month'] = df['Formatted Date'].dt.month
df['Day'] = df['Formatted Date'].dt.day
sns.set(style="whitegrid")

# ----------------------------------------
# PERSON 1 : Temperature Analysis
# ----------------------------------------
# Graph 1: Temperature Distribution
plt.figure(figsize=(8,5))
sns.histplot(df['Temperature (C)'], bins=40, kde=True)
plt.title("Distribution of Temperature")
plt.show()

# Graph 2: Temperature vs Humidity
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(x=df['Temperature (C)'], y=df['Humidity'])
plt.title("Temperature vs Humidity")
plt.show()

# Graph 3: Temperature Trend Across Months
plt.figure(figsize=(8,5))
sns.lineplot(x=df['Month'], y=df['Temperature (C)'])
plt.title("Temperature Trend Across Months")
plt.show()
# ----------------------------------------
# PERSON 2 : Humidity Analysis
# ----------------------------------------
# Graph 4: Humidity Distribution
plt.figure(figsize=(8,5))
sns.histplot(df['Humidity'], bins=30, kde=True)
plt.title("Humidity Distribution")
plt.show()

# Graph 5: Humidity vs Weather Summary
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Summary'], y=df['Humidity'])
plt.xticks(rotation=90)
plt.title("Humidity vs Weather Summary")
plt.show()

# Graph 6: Humidity vs Visibility
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(x=df['Humidity'], y=df['Visibility (km)'])
plt.title("Humidity vs Visibility")
plt.show()
# ----------------------------------------
# PERSON 3 : Wind Analysis
# ----------------------------------------

# Graph 7: Wind Speed Distribution
plt.figure(figsize=(8,5))
sns.histplot(df['Wind Speed (km/h)'], bins=40)
plt.title("Wind Speed Distribution")
plt.show()

# Graph 8: Wind Speed vs Temperature
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['Wind Speed (km/h)'], y=df['Temperature (C)'])
plt.title("Wind Speed vs Temperature")
plt.show()

# Graph 9: Wind Speed Outliers
plt.figure(figsize=(8,5))
sns.boxplot(y=df['Wind Speed (km/h)'])
plt.title("Wind Speed Outliers")
plt.show()
# ----------------------------------------
# PERSON 4 : Pressure & Visibility
# ----------------------------------------
```

```
# Graph 10: Pressure Distribution
plt.figure(figsize=(8,5))
sns.histplot(df['Pressure (millibars)'], bins=40)
plt.title("Pressure Distribution")
plt.show()

# Graph 11: Pressure vs Visibility
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['Pressure (millibars)'], y=df['Visibility (km)'])
plt.title("Pressure vs Visibility")
plt.show()

# Graph 12: Pressure Trend Across Months
plt.figure(figsize=(8,5))
sns.lineplot(x=df['Month'], y=df['Pressure (millibars)'])
plt.title("Pressure Trend Across Months")
plt.show()


# ----------------------------------------
# PERSON 5 : Correlation & Weather Type
# ----------------------------------------

# Graph 13: Correlation Heatmap
plt.figure(figsize=(10,6))
corr = df[['Temperature (C)','Humidity','Wind Speed (km/h)',
```

```
        'Pressure (millibars)','Visibility (km)']].corr()

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

# Graph 14: Precipitation Type Count
plt.figure(figsize=(8,5))
sns.countplot(x=df['Precip Type'])
plt.title("Precipitation Type Count")
plt.show()

# Graph 15: Temperature vs Precipitation Type
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Precip Type'], y=df['Temperature (C)'])
plt.title("Temperature vs Precipitation Type")
plt.show()
```